

FitBuddy: AI-Powered Personal Fitness Companion

Comprehensive Project Planning Phase Documentation

Project Name: FitBuddy Mobile App

Prepared By: Product & Tech Planning Team

Phase: Phase 1 - Project Planning & Feasibility

Document Version: 1.0.0

Target Launch: Q3 2026

Status: Approved

1. Executive Summary & Vision

FitBuddy is an all-in-one smart mobile application designed to democratize personalized fitness coaching and nutrition tracking. By leveraging real-time computer vision for form feedback, dynamic workout generation, and AI-driven meal analysis, FitBuddy fills the gap between static fitness apps and costly personal trainers.

Core Vision Statement

To empower individuals of all fitness levels to achieve sustainable health goals through intelligent, accessible, and adaptive real-time guidance.

2. Project Objectives & Scope

2.1 Core Objectives

- **User Engagement:** Reach 50,000 active monthly users (MAU) within 6 months of release.
- **AI Form Guidance:** Deliver accurate computer-vision posture correction with <200ms latency.
- **Customization:** Provide hyper-personalized daily workout and nutrition recommendations based on biometrics, goals, and equipment availability.

2.2 Scope Definition

In-Scope (Phase 1 Release)

- iOS & Android mobile apps (React Native)
- User onboarding, goal setup, & biometric profiling
- AI real-time posture check (Camera-based)
- Custom workout planner & logging system
- Calorie & macronutrient intake tracker
- Wearable integration (Apple Health, Google Fit)

Out-of-Scope (Future Iterations)

- 1-on-1 video consultations with certified coaches
- E-commerce store for gym gear and supplements
- Desktop / Web application portal
- Custom hardware/wearable development

3. Work Breakdown Structure (WBS) & Timeline

The project will execute across 6 distinct phases over a 24-week delivery roadmap:

Phase	Milestone / Stage	Duration	Key Deliverables
Phase 1	Project Planning & Scope	Weeks 1 - 3	Requirements doc, Tech Architecture, UI/UX Wireframes
Phase 2	UI/UX Design & Prototyping	Weeks 4 - 7	Figma Design System, High-Fi Prototypes, User Testing
Phase 3	Core Engine & Backend Dev	Weeks 8 - 14	REST APIs, Auth, Database Schemas, Wearable Sync
Phase 4	AI Engine & Mobile Frontend	Weeks 12 - 19	Pose Estimation Model, React Native UI, Workout Engine
Phase 5	QA, Security & Performance	Weeks 20 - 22	End-to-End Testing, Security Audit, Beta Release
Phase 6	Deployment & Growth	Weeks 23 - 24	App Store & Play Store Launch, Analytics Setup

4. System Architecture & Tech Stack

Component	Selected Technology	Rationale
Frontend Framework	React Native / Expo	Cross-platform code sharing (iOS/Android) with native performance.
Backend Service	Node.js (TypeScript) + NestJS	Scalable, strongly-typed asynchronous backend architecture.
Database	PostgreSQL + Prisma ORM	Relational integrity for user data, workouts, and transaction logs.
AI / Computer Vision	MediaPipe / TensorFlow Lite	On-device real-time pose tracking without sending video frames to server.
Cloud & Hosting	AWS (ECS, Lambda, S3)	Enterprise-grade infrastructure, elasticity, and compliance support.

5. Risk Management Plan

Risk Event	Impact	Probability	Mitigation Strategy
Camera Pose Tracking Latency / Overheating	HIGH	MEDIUM	Optimize frame processing rate (15 FPS target) and run pose estimation on quantized lightweight model on-device.
Health Data Privacy Compliance (HIPAA/GDPR)	HIGH	LOW	Implement end-to-end encryption for health metrics, local storage for sensitive logs, and strict consent flow.
Scope Creep during MVP Development	MEDIUM	HIGH	Strict adherence to Phase 1 product requirements doc (PRD) with bi-weekly change control committee reviews.

6. Resource & Budget Allocation

- **Engineering Team (6):** 1 Lead Architect, 2 Mobile Devs, 2 Backend Devs, 1 ML Engineer
- **Design Team (2):** 1 Product Designer (UI/UX), 1 Motion/Graphic Specialist
- **Quality Assurance (2):** 1 QA Lead, 1 Automation Specialist
- **Project Management (1):** Technical Product Manager